

Repair Request UI / Screen Flow / Wireframe

Responsive Angular UX + Performance Contract

Document Control	Value
Document ID	RR-UI-001
Version	1.2 - Performance Revision Candidate
Status	Ready for Technical Baseline Approval
Revision Date	12 September 2026
Supersedes	RR-UI-001 v1.1
Revision Scope	Add Angular request/render/bundle/report performance rules while preserving screen flow, role/state visibility and API mappings.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

Repair Request UI / Screen Flow / Wireframe

Responsive Web UX Technical Design / MVP Baseline Revision

Document Control	Value
Document ID	RR-UI-001
Version	1.1 - Technical Design Baseline Candidate
Status	Ready for Technical Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	Requirement v1.5; BR v1.5; State v1.5; UC v1.6; DD v1.3; Test v1.2; ERD/DB/API v1.1
Audience	UX/UI, SA, Frontend Developer, Backend Developer, QA, Security Reviewer
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	กำหนด screen inventory, navigation, role/state-aware actions, validation/error UX, responsive behavior and API mapping.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. UI Design Principles

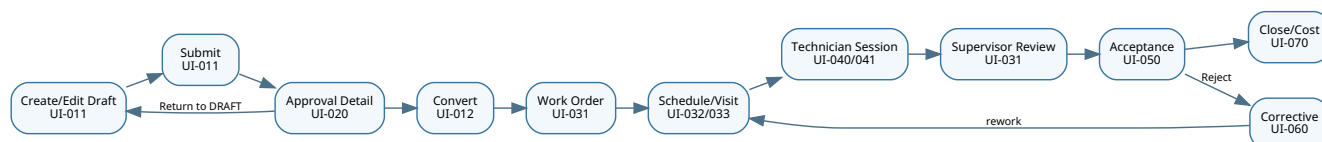
Area	UI Contract
Action-driven state	No direct status editing. Buttons call explicit command APIs only.
Role + scope	Menu/list/detail/actions filtered by role/tenant/site/ownership/assignment; backend remains authority.
State aware actions	Show only actions meaningful for current state; backend validates again.
Concurrency	Keep ETag; 409 shows “data changed” and reload-before-retry.
SLA	Show started/due/effective due/riskAt/breached/stopped. Risk indicator begins T-4h; breach remains visible after override.
Audit	Timeline read-only; no edit/delete.
Files	Show type/size/scan; PENDING processing; FAILED rejected; CLEAN usable.
Responsive	Desktop-first; technician and acceptance flows task-first mobile.
Notifications	Show only current user’s in-app notifications; event routing follows D-14; failed delivery does not make business action look failed.

2. Global Navigation / Screen Inventory

ID	Screen	Primary Role	Main Purpose
UI-001	Dashboard	All authorized	My work, action required, SLA risk/breach cards, notification links
UI-010	Repair Request List	Requester/Approver/Coordinator	Search/filter/list/new Request
UI-011	Request Create/Edit	Requester	Save Draft / Submit / attachment
UI-012	Request Detail	Authorized	Lifecycle, current actions, approval/timeline
UI-020	Approval Inbox/Detail	Approver	Approve / Reject / Return
UI-030	Work Order List	Operational roles	Search/filter Work Orders
UI-031	Work Order Detail	Coordinator/Team Lead/Supervisor	Overview/Visits/Summary/Cost/Acceptance/Timeline
UI-032	Visit Scheduler	Coordinator	Schedule/assign
UI-033	Visit Management	Coordinator	Reassign/reschedule/cancel/MISSED decision
UI-040	My Visits	Assigned Technician	Mobile task list
UI-041	Work Session	Assigned Technician	Check-in/Pause/Resume/Check-out/evidence
UI-050	Customer Acceptance	Designated Contact	Summary/evidence; Accept/Reject
UI-060	Corrective Action	Team Lead/Supervisor/Coordinator	Plan/approve/schedule rework
UI-070	Cost Summary	Team Lead/Supervisor	Prepare/review/close guard
UI-080	Time Correction	Tech/Team Lead/Supervisor	Request/decide correction

ID	Screen	Primary Role	Main Purpose
UI-090	SLA / Reports	Supervisor/Coordinator/reporting	SLA status Risk/Breach/Override; reports/export
UI-100	Timeline / Audit	Authorized	Read-only audit/timeline
UI-110	Notifications	Authenticated	Own in-system notification inbox/read status

3. End-to-End Screen Flow



UI flow mapped to explicit API/state actions. Return is an action to DRAFT; corrective rework loops to the same designated contact.

4. Screen-to-API Mapping

Screen	Read APIs	Command APIs	Concurrency
Request List/Detail	RR-API-003/004; AUD-API-001	RR-API-001/002/005..010; FILE-API-*	Request ETag
Approval	RR-API-004	RR-API-006/007/008	Request ETag
Work Order	WO-API-001; SLA-API-001; AUD-API-001	WO-API-002..011; CST-API-*	WO/Visit ETag
Technician Session	WO/Visit/Session detail	WS-API-001..004	Visit/Session ETag
Acceptance	WO detail	ACC-API-001/002	WO ETag
Corrective	WO + corrective DTO	CA-API-001..003	CA/WO ETag
Time Correction	Session + corrections	TIME-API-001..003	Correction state/token
SLA/Reports	SLA-API-001; REP-API-001	SLA-API-002	SLA token
Notifications	NTF-API-001	NTF-API-002	n/a

5. Validation / Error / Confirmation UX

Condition	Required UX	API / State Consequence
Mandatory/master invalid	Inline field error + summary; retain input	422; no transition
Duplicate warning	Show matching warning; continuation reason required	Stay DRAFT until reason; then submit
Unauthorized/out of scope	Generic access/not found; do not expose hidden data	403/404 non-leaking
Stale rowVersion	Banner; reload latest before retry	409 concurrency conflict; no partial transition
Illegal state	Disable obvious invalid action; refresh on server conflict	409 state conflict
Notification send failed		

Condition	Required UX	API / State Consequence
	Business success stays success; optional delivery warning only	Business state committed; retry recipient event
File PENDING	Processing badge; cannot satisfy required evidence	No guarded transition
File FAILED	Rejected badge + replace	Cannot satisfy evidence
Routing failure	Request stays SUBMITTED; authorized staff sees routing issue	No auto approve
SLA Risk/Breach	Risk at T-4h; breach persistent even after later override/correction	Risk event / sticky breach
MISSED decision	Original Visit shown MISSED immutable; decision creates linked new Visit or No follow-up	Original state remains MISSED

6. Security / Privacy UX Rules

Control	UI Requirement
Tenant/Site	Do not let user choose tenant as authority; Site options only from backend scope.
IDOR	Every deep link handles non-leaking 403/404.
PII	Acceptance contact email/phone only to authorized role; not in client logs.
Attachments	No public object URL; authorized download API only.
Reason fields	Cancel/Reject/Return/Reassign/Reschedule/Override/Correction mandatory as rules require.
Audit	Read-only timeline; no edit/delete.
Client caching	Avoid persistent browser storage of Restricted PII/file URLs unless security architecture approves.

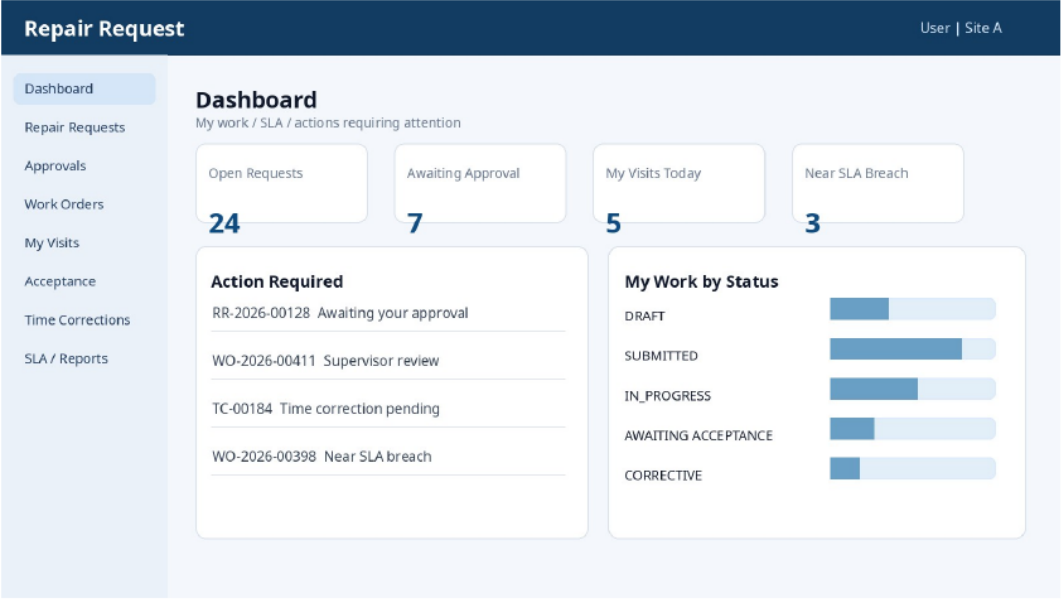
7. Responsive Behavior

Breakpoint / Flow	Behavior
Desktop	Side nav + split detail/action; server-side tables.
Tablet	Collapse nav; action panel below summary; preserve state/SLA header.
Mobile - Technician	Current Visit first; check-in/pause/resume/check-out/evidence; minimal typing.
Mobile - Acceptance	Summary/evidence first; Accept/Reject prominent; Reject reason inline.
Long tables	Card/list summary on small screens; avoid horizontal scroll for primary actions.

8. Reference Wireframes

ภาพต่อไปนี้นำมาจาก wireframe เดิมเพื่อคง visual intent; revised business rules and API mappings in this document supersede any conflicting text label in the old illustration.

5.1 Dashboard + Repair Request List



5.2 Request Form + Approval Detail

Repair Request

User | Site A

Dashboard

Repair Requests

Approvals

Work Orders

My Visits

Acceptance

Time Corrections

SLA / Reports

Repair Requests - Create / Edit

DRAFT only; mandatory validation enforced on Submit

Site *

Plant A

Category *

Machine

Priority *

High

Equipment

Mixer MX-12

Location

Mixing Area 2

Contact *

Somchai / 08x...

Preferred Start *

12 Sep 09:00

Preferred End *

12 Sep 11:00

Problem / Description *

Motor has abnormal vibration and noise.

Save Draft

Submit

Repair Request

User | Site A

Dashboard

Repair Requests

Approvals

Work Orders

My Visits

Acceptance

Time Corrections

SLA / Reports

Approvals - Request Detail

Action panel depends on role + state + scope

RR-2026-00128

UNDER_REVIEW | High | Plant A

Category

Machine

Equipment

Mixer MX-12

Problem

Abnormal vibration and noise

Requester

N. User

Submitted

11 Sep 20:10 UTC

SLA Due

12 Sep 20:10 UTC

Available Actions

Approve

Reject

Return for Correction

Reason required for Reject / Return

Repair Request - UI / Screen Flow / Wireframe

Reference wireframe - original page 7; behavior governed by UI v1.1 revised rules.

RR-UI-001 | Repair Request Baseline Revision

Page 6 / 14

5.3 Work Order Detail + Schedule Visit

Repair Request

User | Site A

Dashboard

Repair Requests

Approvals

Work Orders

My Visits

Acceptance

Time Corrections

SLA / Reports

Work Orders - Detail

Aggregate view: request, visits, SLA, summary, acceptance, audit

WO-2026-00411

IN_PROGRESS

Source: RR-2026-00125 | Team: Maintenance A | SLA: 4h 12m left

Overview

Visits

Work Summary

Cost

Acceptance

Timeline

Current Visit

12 Sep 09:00 - 11:00 | Technician: T.001

Status: IN_PROGRESS

View Session

Next Actions

Submit Summary

Cancel Work Order

Actions shown only when guard conditions pass.

Repair Request

User | Site A

Dashboard

Repair Requests

Approvals

Work Orders

My Visits

Acceptance

Time Corrections

SLA / Reports

Work Orders - Schedule Visit

Coordinator

Schedule Service Visit

Assigned Team *

Maintenance A

Technician

T.001

Start *

12 Sep 09:00

End *

12 Sep 11:00

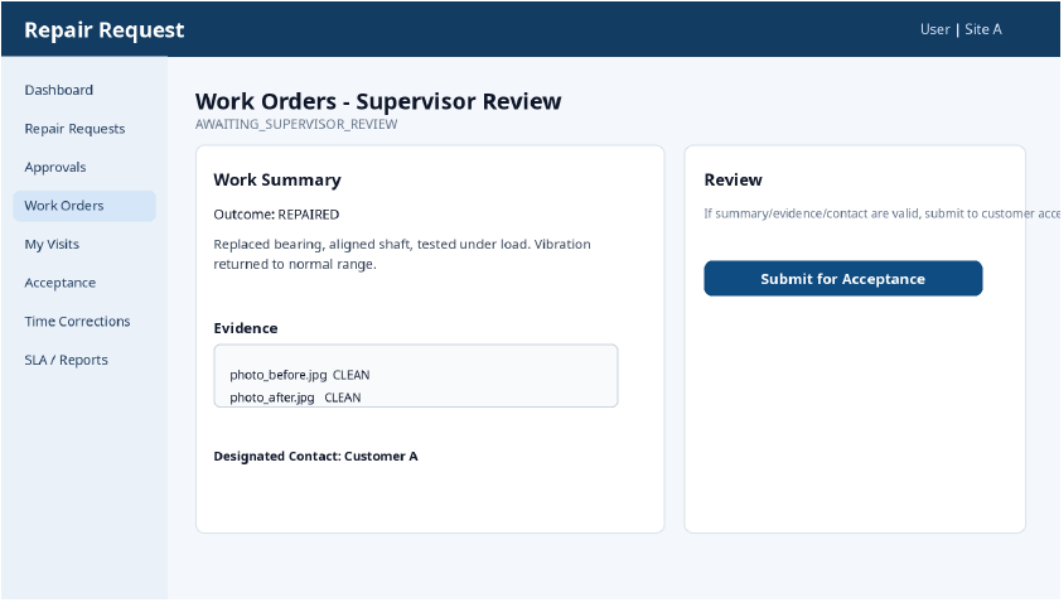
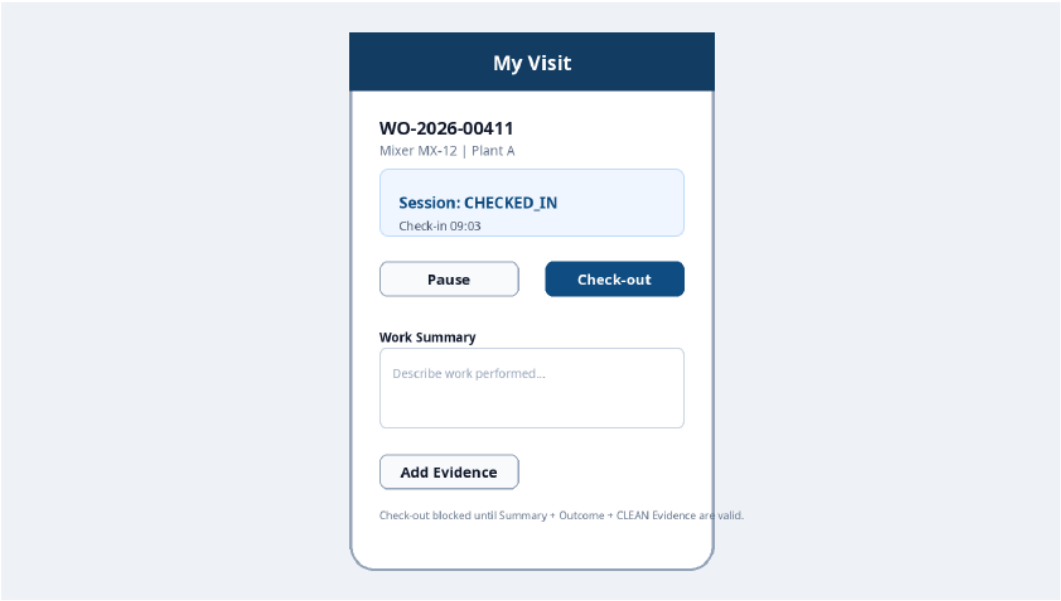
Guard checks

• team/site active and in scope • end >= start • work order must be OPEN for initial schedule

Cancel

Schedule

5.4 Technician Mobile + Supervisor Review



5.5 Customer Acceptance + Corrective Action

Customer Acceptance

WO-2026-00411

Awaiting your decision

Summary

Repair completed. Bearing replaced and machine tested. Evidence attached.

Accept

Reject

Reject reason *

Repair Request

User | Site A

Dashboard

Repair Requests

Approvals

Work Orders

My Visits

Acceptance

Time Corrections

SLA / Reports

Work Orders - Corrective Action

Customer Reject -> Team Lead Plan -> Supervisor Approve -> Rework

Corrective Cycle #1

Status: CORRECTIVE_ACTION_REQUIRED

Corrective Plan *

Replace seal, retest pressure, attach before/after evidence

Plan Evidence *

corrective_plan.pdf CLEAN

Upload

Submit Plan

Approve Plan

Schedule Rework

Repair Request - UI / Screen Flow / Wireframe

Reference wireframe - original page 10; behavior governed by UI v1.1 revised rules.

RR-UI-001 | Repair Request Baseline Revision

Page 9 / 14

5.6 Time Correction + SLA / Reports

Repair Request

User | Site A

Dashboard

Repair Requests

Approvals

Work Orders

My Visits

Acceptance

Time Corrections

SLA / Reports

Time Corrections

Technician / Team Lead request; Supervisor decides

Request Time Correction

Timestamp Type *

CHECK_OUT

Current Value *

12 Sep 10:58 UTC

New Value *

12 Sep 11:12 UTC

Reason *

Forgot to check out immediately after final test.

Submit Request

Supervisor approval updates only the allowed timestamp. SLA is not recalculated.

Repair Request

User | Site A

Dashboard

Repair Requests

Approvals

Work Orders

My Visits

Acceptance

Time Corrections

SLA / Reports

SLA / Reports

Role/site scoped reporting and operational monitoring

From

01 Sep

To

30 Sep

Site

Plant A

SLA

All

Run Report

SLA Summary

Open: 36

Near breach: 5

Breached: 7

Stopped: 18

Report Areas

• Work by status

• Near breach / breached

• By Site / team

• Cancel / Reject history

Export (scoped)

Repair Request - UI / Screen Flow / Wireframe

Reference wireframe - original page 11; behavior governed by UI v1.1 revised rules.

RR-UI-001 | Repair Request Baseline Revision

Page 10 / 14

6. UI States, Validation and Error Handling

Condition	Required UX behavior	API / state consequence
Mandatory / master invalid	Inline field error + validation summary; retain user input	422 VALIDATION_FAILED; no transition
Duplicate warning	Show matching-request warning; require continuation reason before Submit	Request stays DRAFT until reason; then submit
Unauthorized / out of scope	Do not expose hidden data; show generic access/not-found message	403/404 non-leaking behavior
Stale rowVersion	Banner: Data changed by another user. Reload latest before retry.	409 CONCURRENCY_CONFLICT; no partial transition
Illegal state	Disable obvious invalid action; if state changed server-side, show conflict and refresh	409 STATE_CONFLICT
Notification send failed	Do not present business action as failed if transition committed; show success + optional delivery warning only if product chooses	Business state remains committed
File PENDING scan	Show Processing badge; cannot use as mandatory evidence	No check-out / plan submit if required CLEAN evidence missing
File FAILED scan	Show Rejected badge + remove/replace action	Cannot satisfy evidence requirement
Routing failure	Request remains SUBMITTED; show routing issue to authorized staff; no auto-approve	Audit/notification created
SLA breached	Persistent breached indicator even after later override/time correction	breach history is sticky

6.1 Confirmation dialogs

Action	Confirm content
Cancel Request / Visit / Work Order	Reason mandatory; explain terminal/propagation effect
Reject Request / Work	Reason mandatory; customer reject explains Corrective Action creation
Return for Correction	Reason mandatory; explain Request returns to DRAFT
Close Work Order	Confirm summary + cost reviewed; Closed has no Reopen in MVP
SLA Override	Show original due, new effective due, reason and permission warning
Approve Time Correction	Show old vs new timestamp; remind SLA is not recalculated

7. Screen-to-API Mapping

Screen	Read APIs	Command APIs	Concurrency token
Repair Request List	GET /repair-requests	POST /repair-requests	-
Repair Request Detail	GET /repair-requests/{id}; timeline	PATCH draft; submit; approve/reject/return/cancel/con vert	Request ETag on mutable actions
Work Order Detail	GET /work-orders/{id}; SLA; timeline	schedule; submit summary; submit acceptance; close/cancel	Work Order ETag
Visit Scheduler	Work Order detail / visit DTO	schedule/reassign/reschedule/ cancel/missed/follow-up	Visit/WO ETag as applicable
Technician Session	Work Order/Visit/Session detail	check-in/pause/resume/check- out	Session/Visit ETag
Acceptance	Work Order detail	accept/reject	Work Order ETag
Corrective Action	Work Order + corrective DTO	submit plan/approve/schedule rework	Corrective/WO ETag
Time Correction	Session + corrections	request/approve/reject	Correction current state
SLA / Reports	SLA record; report query	SLA override	SLA current record

Route names ใน UI เป็น technical design convention. API path exact shape ปีค RR-API-001 v1.0; UI ต้องไม่สร้าง generic state update call.

8. Security / Privacy UX Rules

Control	UI requirement
Tenant / Site scope	ไม่ให้ user เลือก tenant เป็น authority; Site option แสดงเฉพาะ scope ที่ backend คืนให้
IDOR	ห้ามสร้าง predictable navigation ที่ถือว่า ID เท่ากับสิทธิ์; ทุก deep link ต้องรับมือ 403/404 แบบไม่ leak
PII	Email/phone ของ Acceptance Contact แสดงเฉพาะ role ที่มีสิทธิ์ และไม่ฝังใน client logs
Attachment / Evidence	ไม่มี public object URL; download ผ่าน authorized API; แสดง metadata เท่าที่จำเป็น
Reason fields	Cancel/Reject/Return/Reassign/Reschedule/Override/Correction แสดง mandatory reason ตาม business rule
Audit	Timeline read-only; ไม่มี Edit/Delete audit action
Admin	Admin screens สำหรับ master/SLA policy ต้องแยก permission; Administrator ไม่ impersonate business owner โดยค่าเริ่มต้น
Client caching	หลีกเลี่ยงเก็บ Restricted PII/file URLs ใน persistent browser storage เว้นแต่ architecture security อนุมัติ

8.1 Responsive behavior

Breakpoint intent	Behavior
Desktop	Side navigation + split detail/action panels; tables with server-side paging/filtering
Tablet	Collapse side nav; action panel below summary; preserve status + SLA header
Mobile - Technician	Task-first: current visit, check-in/pause/resume/check-out, evidence upload; minimum typing
Mobile - Acceptance	Summary/evidence first; Accept/Reject prominent; Reject reason inline
Long tables	Switch to card/list summary on small screens; avoid horizontal scrolling for primary actions

9. Design Decisions / Deferred Production Items

Item	Status / Impact
Authentication/login mechanism	Deferred to Architecture/Security; authorization semantics already locked.

Item	Status / Impact
Master-data CRUD details	Deferred to Master Data extension; source does not define full master schemas.
Exact export format / max page size	Deployment/product configuration.
Brand colors/icons	Illustrative, not business semantics.
Provider-specific notification center delivery	Operations/architecture refinement; event/recipient business semantics are locked.

10. UI Readiness

Core MVP screens map to revised Use Case, State and API IDs with no known blocking UI flow gap. Ready for frontend component design and Sprint planning after technical baseline approval.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อความขัดกัน โดยไม่เปลี่ยน Business Baseline.

11. Angular Frontend Performance Contract

Area	Frontend Technical Rule	Boundary
List/Search	Use server-side filter/sort/paging. Search/filter streams should cancel superseded reads (e.g., switchMap/AbortSignal pattern) and may debounce free-text input to avoid duplicate requests.	Debounce interval is technical UX configuration, not a business rule.
Route loading	Feature-oriented Angular architecture may lazy-load routes/features to reduce initial bundle when practical.	Lazy loading is recommended technical design, not a functional acceptance requirement by itself.
Rendering	Use stable identity/track strategy for repeated rows. Virtual scrolling is introduced only when profiling shows large DOM/render cost.	Do not force virtualization for small bounded server-side pages.
State / subscriptions	Prefer feature-local state/signals/RxJS and deterministic teardown; avoid duplicate subscriptions/API calls.	Global store is added only when cross-feature complexity justifies it.
Files	Upload/download UI remains responsive and shows PENDING/FAILED/CLEAN states; browser must not persist restricted file URLs/PII as a performance shortcut.	Security rules have priority over caching.
Errors / concurrency	409 reload-before-retry behavior remains unchanged; optimization may not auto-repeat stale state-changing commands.	No hidden retry that can duplicate business actions.

12. Frontend Performance Verification

For critical list/search screens record production-build bundle/network evidence, duplicate/cancelled request behavior, render timing sample and API correlation. No bundle-size or render-time release threshold is invented until a stable measured baseline is approved.

PERF-DEC-01 - Performance Measurement Statistic (Pending Approval).

Business Baseline ระบุ “Normal List/Search <= 3 seconds at 100 concurrent users and 100,000 records per Site” แต่ไม่ได้ระบุว่า 3 seconds หมายถึง Average, Maximum, p95 หรือ p99. จึง **ห้ามถือ p95 เป็น Requirement ที่อนุมัติแล้ว**. Recommendation สำหรับการอนุมัติคือ p95 end-to-end API response time <= 3.0 seconds; ก่อนอนุมัติ ให้บันทึก Average/p50/p95/p99/throughput/error rate ครบและรายงานผลเป็น diagnostic evidence โดยยังไม่ประกาศ Pass/Fail ของ release gate จาก percentile ใด percentile หนึ่ง.

13. Report / Export UX Boundary

Exact export format/max page size remains product/deployment configuration. If measured export workload is large enough to affect interactive API or browser memory, asynchronous export can be proposed as a Technical Design change; it is not silently added as an MVP business requirement.